

Student Name:

Reg/ID #:

Consider the following RISC-V assembly code:

```
add  x13, x14, x15  # instruction address is 0x100
addi x10, x14, 25
ld   x12, 16(x4)
beq  x12, x13, label # label is 5 instructions forward
```

For each of the instructions, write values of the control and data signals. You can use don't care or x if needed. Assume x1 contains 1, x2 2, x3 3 and so on

Signal	add x13, x14, x15	addi x10, x14, 25	ld x12, 16(x4)	beq x12, x13, label
RegWrite	1	1	1	0
ALUSrc	0	1	1	0
PCSrc	0	0	0	0
MemRead	0	0	1	0
MemWrite	0	0	0	0
MemtoReg	0	0	1	0
rs1	01110	01110	00100	01100
rs2	01111	-	-	01101
rd	01101	01010	01100	-
ALU operand # 1	14	14	4	12
ALU operand # 2	15	25	16	29
Write data	29	39	\$(20)	-
PC	0x100	0x104	0x108	0x10C